

**Manchester
Metropolitan
University**

Course Notes

Maths for Data Science

6G5Z0050

Semester 2 - 2025-2026

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Probability

Key terms

KSBs: K13, S11, S53, B1

Probability, independence, joint and union probabilities, conditional probability, Bayes' Theorem

Probability is the mathematical study of uncertainty - when the outcome of events is unknown, we can use it to compare the likelihood of different outcomes and use this information to make decisions. As such, it's fundamental to the study of statistics and data science.

1.1 Basic concepts

Definition 1.1 - Experiments, Outcomes and Events

We use the word **experiment** to describe an occurrence we don't know the result of yet.

An experiment may have multiple possible **outcomes**, each of which is one possible result of the experiment. We use the word **event** to describe a collection of possible outcomes.

Example 1.1.

An experiment involves rolling a fair six-sided die. It will have an **equal probability of landing with any one of its six sides facing up.**

We might talk about the event of 'rolling an odd number' - this encompasses the outcomes of **rolling a 1, 3 or 5.**

We write the probability of an outcome X using the notation $P(X)$, and probabilities can be expressed as a percentage **(70%)** or as a decimal **(0.7)**. They always take values between **0 and 1** or **0% and 100%.**

Definition 1.2 - Negative probability

If the probability of an event X happening is $P(X)$, the probability of it not happening is **$1 - P(X)$ or $100\% - P(X)$.**

To calculate the probability of an event, we could write down a **possibility space**, listing all the possible outcomes and counting how many would give us the outcome we need.

Example 1.2. If I'm rolling a six-sided fair die, there are **Six possible outcomes:**



Since these outcomes are **all equally likely**, we know the probability of each is $\frac{1}{6}$. The event **"rolling an odd number"** corresponds to a subset of the possible outcomes of size **3**:



The probability of this event is $\frac{3}{6} = \frac{1}{2}$.

Example 1.3.

Let's consider a more complicated example, involving rolling a six-sided die and flipping a fair coin. The possible combinations of outcomes are shown in this table:

H	H	H	H	H	H
T	T	T	T	T	T

>4

There are **6** possible outcomes for the die, and **2** for the coin. The total number of possible outcomes is $6 \times 2 = 12$. If we consider the event **"rolling a number greater than 4"** this corresponds to the **four outcomes at the right hand side of the table.**

Theorem 1.1 - Probability from a Possibility Space

The probability of an event E is given by:

$$P(E) = \frac{\text{the number of ways } E \text{ can occur}}{\text{the total number of possible outcomes.}}$$

Example 1.4. Let's consider the same example as before - rolling a six-sided die and flipping a fair coin.

- The event $A = \text{'rolling a number greater than four'}$ includes 4 of the 12 outcomes

$$\text{so } P(A) = \frac{4}{12} = \frac{1}{3} = 33.\bar{3}\% = 0.\bar{3}$$

- The event $B = \text{'rolling an odd number'}$ includes 6 of the 12 outcomes

$$\text{so } P(B) = \frac{6}{12} = \frac{1}{2} = 50\% = 0.5$$

- The event $C = \text{'rolling an odd number and getting heads'}$ includes 3 of the 12 outcomes.

$$\text{so } P(C) = \frac{3}{12} = \frac{1}{4} = 25\% = 0.25$$

Exercise 1.1.

- a) A spinner has eight sections, each of which is equally likely to land on the table when the spinner is used. The sections are numbered 1-8. What is the probability that the spinner will land on a square number?
- b) A set of 100 Scrabble tiles, with the distribution shown, is placed in a bag, and one tile is drawn out. What is the probability it is a vowel?
- 12×[E] 9×[A I] 8×[O] 6×[N R T] 4×[D L S U] 3×[G] 2×[B C F H M P V W Y] 1×[J K Q X Z]
- c) In the UK lottery, six balls are drawn from a set numbered 1-59 in the machine, which is carefully designed to ensure each ball has an equal probability of being drawn. What is the probability that the first ball drawn is an odd number?
- d) A lecturer surveyed the 29 students in their class about how they travelled to university. 5 students walked, 7 students cycled, 6 students travelled by bus, and the rest of the students in the class travelled by car. A student is picked at random from the class. Find the probability that the student travelled to university either by bus or by car.

1.2 Independent events

Definition 1.3 - Independent Events

Two events are **independent** if *the probability of either event is not affected by the outcome of the other.*
Otherwise, we say the events are *dependent.*

Definition 1.4 - Mutually exclusive events

Two events are **mutually exclusive** if *they cannot both happen at the same time.*
If a single experiment has multiple possible outcomes, the outcomes will be *mutually exclusive,*
as only one of them can happen.

Example 1.5.

- If we flip a fair coin, the outcomes 'the coin lands heads' and 'the coin lands tails' *are mutually exclusive.*
- If we flip a fair coin twice, the outcome of the second coin flip *is independent of the outcome of the first.*
- If we have a bag containing half red and half blue balls, the colour of the second ball drawn from the bag being red *is dependent on whether the first ball drawn was red or blue.*

We may not know whether two events are independent, and might seek to establish this by comparing their probabilities. As we will see, the independence of events can affect the probabilities of them happening together.

Exercise 1.2. Are the following events dependent or independent?

- Ali spins a spinner once a day for a week. A = 'on Tuesday, it lands on 1'; B = 'on Friday, it lands on 3'
- A playing card is drawn from a deck of cards. A = 'the card is a club'; B = 'the card is a 7'.
- A playing card is drawn from a deck of cards. A = 'the card is a club'; B = 'the card is black'.
- A random MMU student is chosen. A = 'the student is taller than the median height'; B = 'the student is a member of the MMU basketball team'

Exercise 1.3. Which of the following are mutually exclusive?

- A playing card drawn from a deck is red; it is black
- A playing card drawn from a deck is red; it is a club
- A playing card drawn from a deck is red; it is a diamond
- A die is rolled and the result is a 6; the result is a 5
- A die is rolled and the result is a 6; the result is odd
- The maximum temperature tomorrow will be 5°C ; the minimum temperature tomorrow will be 6°C
- A participant in a study has never eaten chips; they eat chips at least once a week
- A participant in a study has never eaten chips; they have never been to the gym

1.3 Combining probabilities

When considering multiple events, we can combine their probabilities to determine the probability of multiple events happening. Most commonly we consider the probability of two events both happening (A and B) or the probability of one or the other of two events happening (A or B).

1.3.1 Joint Probability

Definition 1.5 - Joint Probability

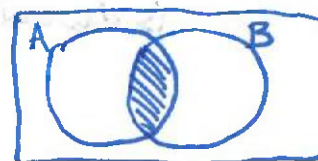
The probability of two independent events both happening is the product of the probabilities of each of the events.

Indeed, let A and B be two independent events, with probabilities $P(A)$ and $P(B)$ respectively. Then the probability of both A and B happening, denoted $P(A \cap B)$, is given by

$$P(A \cap B) = P(A) \times P(B)$$

Note

We use the symbol $\cap$ to mean 'AND'. This mirrors its use in mathematical logic to describe two events both happening and in describing the intersection of two sets.



Example 1.6.

Considering the same possibility space from earlier:

- Let A = 'rolling an odd number' and B = 'flipping the coin and getting heads'. Then $P(A) = \frac{6}{12} = \frac{1}{2}$ and $P(B) = \frac{1}{2}$. These are independent, so $P(A \cap B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.
From our earlier example, by counting possibilities, we also found $\frac{3}{12} = \frac{1}{4}$ was $P(A \cap B)$.
- For probabilities expressed as a percentage, we need to use the decimal form.
 $P(A) \times P(B) = 50\% \times 50\% = 0.5 \times 0.5 = 0.25 = 25\%$.
- Let A = 'rolling a four' and B = 'flipping the coin and getting tails'. Then $P(A) = \frac{2}{12} = \frac{1}{6}$ and $P(B) = \frac{1}{2}$. These are independent, so $P(A) \times P(B) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12} = P(A \cap B)$.
This joint probability corresponds to exactly one of the 12 outcomes listed in the possibility space, so this confirms $P(A \cap B) = \frac{1}{12}$.

Note this only works for independent events. If the events are not independent, we will need to consider the probabilities more carefully, and later we will see ways to combine dependent probabilities.

Example 1.7.

A bag contains 5 red and 10 blue balls. The probability of a ball drawn from the bag being red is $\frac{5}{15} = \frac{1}{3}$ and the probability of a ball drawn from the bag being blue is $\frac{10}{15} = \frac{2}{3}$.

Two balls are drawn from the bag, one at a time, without replacement. Consider the probability of both balls being red. If these events were independent, we could multiply the probability of drawing a red ball by itself - this would give:

$$P(\text{two red balls drawn}) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

However, this is not the correct probability - it is actually $\frac{2}{21}$. Later we will see how to calculate probabilities when events are not independent.

Exercise 1.4.

The probabilities of various events are given, regarding a normal UK Lottery draw.

$$A = \text{'all the balls drawn are numbers greater than 10'} \quad P(A) = 31\%$$

$$B = \text{'the ball set "Guinevere" is chosen for the draw'} \quad P(B) = 25\%$$

$$C = \text{'the jackpot for the draw is more than £10m'} \quad P(C) = 17\%$$

$$D = \text{'more than the median number of tickets are sold'} \quad P(D) = 50\%$$

For each of the joint probabilities listed below, assuming the events are independent, calculate the joint probability. Discuss whether all of these are independent, and what information you would need in order to determine this.

$$i) P(A \cap B) \quad ii) P(A \cap C) \quad iii) P(B \cap D) \quad iv) P(C \cap D)$$

1.3.2 Union Probability

Definition 1.6 - Union Probability for mutually exclusive events

For two mutually exclusive events, the probability of **one or the other** happening is **the sum of the probabilities of each of the events.**

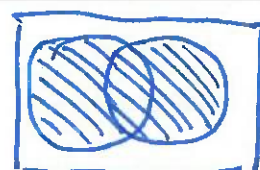
Indeed, let A and B be two mutually exclusive events, with probabilities $P(A)$ and $P(B)$ respectively.

Then the probability of either A or B happening, denoted $P(A \cup B)$ is given by

$$P(A \cup B) = P(A) + P(B)$$

Note

We use the symbol $\cup$ to mean "OR". This mirrors its use in mathematical logic to describe **one of a pair of events happening** and in describing **the union of two sets.**



Example 1.8. Considering the same possibility space from earlier, of a coin flip and dice roll.

- Let A = 'rolling a two' and B = 'rolling a number greater than 4'. Then $P(A) = \frac{2}{12} = \frac{1}{6}$ and $P(B) = \frac{4}{12} = \frac{1}{3}$. These events are mutually exclusive, so we can calculate:

$$P(A \cup B) = P(A) + P(B) = \frac{1}{6} + \frac{1}{3} = \frac{3}{6} = \frac{1}{2}$$

By counting possibilities, we can see that **6 of the 12 outcomes involve A or B.**

- If the events are not mutually exclusive, we cannot just add the probabilities. Indeed, let C = 'rolling a number below 3' (with $P(C) = \frac{1}{3}$) and D = 'coin lands heads' (with $P(D) = \frac{1}{2}$)

These events are **not mutually exclusive**, and occur in the possibility space $\frac{8}{12} = \frac{2}{3}$

If we were to calculate the union probability by adding, we would get $\frac{1}{2} + \frac{1}{3} = \frac{5}{6} \neq \frac{2}{3}$

This is because **the outcomes in which C and D both occur are counted twice.**
In order to get the correct probability, we can **subtract $P(C \cap D)$ in this case, $\frac{1}{6}$.**

Definition 1.7 - Union Probability

For two events, the probability of **one or the other** happening is the **sum of the probabilities of each event, minus the probability of both of them happening.**

Indeed, let A and B be two events with probabilities $P(A)$ and $P(B)$ respectively. Then the probability of either A or B happening, denoted $P(A \cup B)$ is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Note that if two events A and B are mutually exclusive, then $P(A \cap B) = 0$ and so the formula matches the previous version.

Exercise 1.5.

- i) In a class of 30 students, 8 students are in K-Pop Society, 15 students play a sport, and 5 students are both in the K-Pop Society and play a sport. Let A be the event that a student is in the K-Pop Society and let B be the event that a student plays a sport. Find $P(A \cap B)$, and find $P(A \cup B)$.
- ii) Nic sometimes has a dessert after dinner. Nic has a cookie for dessert on average one night in every four, and ice cream for dessert one night in 10. Sometimes, when it's a special occasion (one night in 20), dessert is both a cookie and ice cream. What is the probability that Nic will have dessert tonight?

1.4 Conditional probability

When events are not independent, we can use conditional probability to calculate the probability of them occurring together.

Definition 1.8 - Conditional probability

Conditional probability is a measure of the probability of an event A occurring, given that an event B has already occurred.

We write "the probability of A given B " as $P(A|B)$

Example 1.9.

In an earlier example, we considered a bag with 5 red balls and 10 blue balls, and the probability of drawing two red balls from the bag on two tries.

$$P(\text{first ball red}) = \frac{1}{3}$$

Once the first ball is drawn, if it is red, there are now 4 red balls & 10 blue balls.

$$\text{So } P(\text{second ball red} \mid \text{first ball red}) = \frac{4}{14}$$

If the first ball drawn were blue, there would be 5 red balls and 9 blue balls.

$$P(\text{second ball red} \mid \text{first ball blue}) = \frac{5}{14}$$

Note that the direction of the condition is important: $P(A|B)$ is not necessarily the same as $P(B|A)$.

Example 1.10.

A fair 6-sided dice is being rolled. Let A = 'the probability the dice lands on a 3', and let B = 'the probability the dice lands on an odd number'.

To find $P(A|B)$: given B has happened, it could be a 1, 3 or 5.

$$\text{So } P(A|B) = \frac{1}{3}.$$

To find $P(B|A)$: given A happened, it landed on a 3, which is odd.

$$\text{So } P(B|A) = 1.$$

Example 1.11.

A random MMU student is chosen. A = 'the student is taller than the median height'; B = 'the student is a member of the MMU basketball team'.

Then $P(A|B)$ (the probability that someone is tall given they are on the basketball team) will be high, as being tall is useful in basketball.

while $P(B|A)$ (the probability that someone is on the basketball team given they are tall) will be lower,

*since there are only a small number of students on the team.

Even though these two things aren't the same, there is a relationship between them. If we know $P(A|B)$ and would like to calculate $P(B|A)$, we can use Bayes' Theorem.

1.4.1 Bayes' Theorem**Theorem 1.2 - Bayes' Theorem**

Bayes' theorem states that, for events A and B (with $P(B) \neq 0$),

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Example 1.12.

I receive 100 emails, of which 12 are spam emails. Let A = 'an email I receive is spam', and B = 'the email contains the word "Congratulations" in it'. Of the 100 emails, 10 contain the word "Congratulations", of which 7 are spam.

This means $P(A) = \frac{12}{100}$, $P(B) = \frac{10}{100}$, $P(B|A) = \frac{7}{12}$

By Bayes' theorem,

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{\frac{7}{12} \cdot \frac{12}{100}}{\frac{10}{100}} = \frac{\frac{7}{100}}{\frac{10}{100}} = \frac{7}{10}$$

Therefore if I receive an email with the word "Congratulations",

there is a 70% probability it is spam.

Example 1.13.

There is a diagnostic test for Horrible Disease which is 99% accurate. In the general population, around 3% of people have the disease. A person receives a positive test result. What is the probability that they have the disease?

Let D = 'having Horrible disease', D' = 'not having Horrible disease' and T = 'getting a positive test result'.

$P(D) = 3\%$. Note that $P(D) + P(D') = 100\%$, so $P(D') = 97\%$.

If someone has the disease, the chances of getting a positive test are 99% and if they don't, the chances are 1% (since it might be wrong). So $P(T) = (3\% \times 99\%) + (97\% \times 1\%)$

$$= (0.03 \times 0.99) + (0.97 \times 0.01) = 0.0394 = 3.94\%$$

We can also say that $P(T|D) = 99\%$

By Bayes' theorem,

$$P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T)} = \frac{0.99 \times 0.03}{0.0394} = 0.754 = 75.4\%$$

Even though the test has come back positive, there is only a 75.4% chance of having the disease.

Exercise 1.6.

- a) In a class of 30 students, 8 students are in K-Pop Society, 15 students play a sport, and 5 students are both in the K-Pop Society and play a sport. Let A be the event that a student in the class is in the K-Pop Society and let B be the event that a student in the class plays a sport. Draw a Venn diagram to illustrate this. State $P(A)$, $P(B)$, $P(A|B)$ and $P(B|A)$, and use Bayes' Theorem to check they are related correctly.

- b) A poorly child comes to the doctor, who treats only children with the flu and measles. 90% of poorly children in the area have the flu, while the other 10% have measles. Let F = 'a child has flu' and M = 'a child has measles'. A well-known symptom of measles is a rash, and let R = 'a child has a rash'. Let the probability of having a rash if you have measles be $P(R|M) = 0.95$. A small number of children with the flu also develop a rash, and $P(R|F) = 0.08$. The doctor examines a child, and finds they have a rash. What is the probability that the child has measles?

Function: PROB (E/S)

The PROB function (in Excel and Sheets) can be used to calculate probabilities given a set of numeric values. You will need to give it a list of values and their probabilities, and it can tell you the probability of the result being equal to a value, or in a range.

`=PROB([list of values],[list of probabilities], [value])` will return $P([value])$.

`=PROB([list of values],[list of probabilities], [lower value], [upper value])` will return the probability of being between [lower value] and [upper value].

	A	B	C	D	E	F	G
1							
2		Data	Probability		Result	Formula	
3		1	0.12		0.12	<code>=PROB(B3:B8,C3:C8, 1)</code>	
4		2	0.15		0.1	<code>=PROB(B3:B8,C3:C8,5)</code>	
5		3	0.2		0.26	<code>=PROB(B3:B8,C3:C8,5, 6)</code>	
6		4	0.27		1	<code>=PROB(B3:B8,C3:C8,1, 6)</code>	
7		5	0.1				
8		6	0.16				
9		Total	1				
10							

Exercise 1.7. Given the example table of probabilities above, write down a function that would give the probability of the result being 2 or 3, and write down a function to give the probability of the result being 6.